



EC200A-CN(TA) QuecOpen AT Commands Manual

LTE Standard Module Series

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About the Document

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1 Introduction

This document presents the AT command set supported by Quectel EC200A-CN(TA) module in QuecOpen® solution.

1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals to its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

The AT command set supported by the module is a combination of international standards such as 3GPP TS 27.007, 3GPP TS 27.005 and ITU-T recommendation V.25ter, as well as the AT commands developed by Quectel.

AT commands implemented by EC200A-CN(TA) fall into three categories syntactically: “Basic”, “S Parameter” and “Extended”, as listed below:

- **Basic Command**

These AT commands have the format of **AT<x><n>**, or **AT&<x><n>**, where **<x>** is the command, and **<n>** is/are the argument(s) for that command. An example of this is **ATE<n>**, which tells the DCE (Data Circuit-terminating Equipment) whether received characters should be echoed back to the DTE (Data

Terminal Equipment) according to the value of **<n>**. **<n>** is optional and a default will be used if it is omitted.

● S Parameter Syntax

These AT commands are in the format of **ATS<n>=<m>**, where **<n>** is the index of the **S** register to set, and **<m>** is the value to assign to it.

● Extended Command

These commands can be operated in several modes, as shown in the following table:

Table 1: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of corresponding Write Command and to give information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of a corresponding Write Command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

Multiple commands can be placed on a single line using a semi-colon (;) between commands. In such cases, only the first command should have **AT** prefix. Commands can be in upper or lower case.

Spaces should be ignored when you enter AT commands, except in the following cases:

- Within quoted strings, where spaces are preserved;
- Within an unquoted string or numeric parameter;
- Within an IP address;
- Within the AT command name up to and including a **=**, **?** or **=?**.

On input, at least a carriage return is required. A newline character is ignored so it is permissible to use carriage return/line feed pairs on the input.

If no command is entered after the **AT** token, **OK** will be returned. If an invalid command is entered, **ERROR** will be returned.

Optional parameters, unless explicitly stated, need to be provided up to the last parameter being entered.

1.3. AT Command Responses

When the AT command processor has finished processing a line, it will output **OK**, **ERROR** or **+CME ERROR: <err>** to indicate that it is ready to accept a new command. Solicited information responses are sent before the final **OK**, **ERROR** or **+CME ERROR: <err>**.

Responses will be in the format of:

```
<CR><LF>+CMD1:<parameters><CR><LF>  
<CR><LF>OK<CR><LF>
```

Or

```
<CR><LF><parameters><CR><LF>  
<CR><LF>OK<CR><LF>
```

1.4. Supported Character Sets

The AT command interface of the module uses the GSM character set by default and supports the following character sets:

- GSM format
- UCS-2
- IRA

These character sets are defined in *3GPP TS 27.005*. The character sets affect transmission and reception of SMS and SMS Cell Broadcast Messages, as well as the entry and display of phonebook entries text field.

1.5. AT Command Port

The two USB ports (USB modem port and USB AT port) support AT command communication and data transfer.

1.6. Unsolicited Result Code

As an Unsolicited Result Code and a report message, the URC is not issued as part of the response

related to an executed AT command. A URC is issued by EC200A-CN(TA) without being requested by the TE and it is issued automatically when a certain event occurs. Typical events leading to URCs are incoming calls (**RING**), received short messages, high/low voltage alarm, high/low temperature alarm, etc.

1.7. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about how to use the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendation or suggestions about how you should design a program flow or what status you should set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there exists a correlation among these examples, or that they should be executed in a given sequence.

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2 General Commands

2.1. ATI Display MT Identification Information

This command delivers the MT identification information text.

ATI Display MT Identification Information	
Execution Command ATI	Response Quectel <objectID> Revision: <revision> OK
Maximum Response Time	300 ms
Characteristics	/
Reference <i>V.25ter</i>	

Parameter

<objectID>	String type. Identifier of device type.
<revision>	String type. Identification text of MT firmware version.

Example

```

ATI
Quectel
EC200A-CNTA
Revision: EC200ACNTAR01A03M2G_OCPU

OK
```

2.2. AT+GSN Request International Mobile Equipment Identity (IMEI)

This command requests the International Mobile Equipment Identity (IMEI) number of the ME which permits the user to identify individual ME device. It is identical with **AT+CGSN** in **Chapter 2.3**.

AT+GSN Request International Mobile Equipment Identity (IMEI)	
Test Command AT+GSN=?	Response OK
Execution Command AT+GSN	Response <IMEI> OK
Maximum Response Time	300 ms
Characteristics	/
Reference V.25ter	

Parameter

<IMEI>	String type. IMEI number of the ME.
---------------------	-------------------------------------

NOTE

The IMEI is unique to each ME, so it can be used to identify an ME.

2.3. AT+CGSN Request International Mobile Equipment Identity (IMEI)

This command requests International Mobile Equipment Identity (IMEI) number of the ME. It is identical with **AT+GSN** above.

AT+CGSN Request International Mobile Equipment Identity (IMEI)	
Test Command AT+CGSN=?	Response OK
Execution Command AT+CGSN	Response <IMEI> OK

Maximum Response Time	300 ms
Characteristics	/
Reference	
3GPP TS 27.007	

Parameter

<IMEI>	String type. IMEI number of the ME.
--------	-------------------------------------

NOTE

The IMEI is unique to each ME, so it can be used to identify an ME.

2.4. ATE Set Command Echo Mode

This command controls whether TA echoes characters received from TE or not during AT command mode.

ATE Set Command Echo Mode	
Execution Command ATE<value>	Response OK
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.
Reference	
V.25ter	

Parameter

<value>	Integer type. Whether to echo the characters received from TE.
0	OFF
1	ON

2.5. AT+CFUN Set MT Functionality

This command selects the level of functionality **<fun>** in the MT. It can also reset the MT.

AT+CFUN Set MT Functionality	
Test Command AT+CFUN=?	Response +CFUN: (list of supported <fun> s),(list of supported <rst> s) OK
Read Command AT+CFUN?	Response +CFUN: <fun> OK
Write Command AT+CFUN=<fun>[,<rst>]	Response OK If there is any error related to MT functionality: +CME ERROR: <err> Or ERROR
Maximum Response Time	15 s, determined by the network.
Characteristics	/
Reference	
3GPP TS 27.007	

Parameter

<fun>	Integer type. 0 Minimum functionality <u>1</u> Full functionality 3 Disable MT from receiving RF signals 4 Disable MT from both transmitting and receiving RF signals 5 Disable (U)SIM 6 Second RX off
<rst>	Integer type. <u>0</u> Do not reset the MT before setting it to <fun> power level. 1 Reset the MT before setting it to <fun> power level. The device is fully functional after the reset. This value is available only for <fun>=1 .
<err>	MT error result code. See Chapter 8.2 for details.

Example

```
AT+CFUN=0 //Switch UE to minimum functionality.
OK
AT+CPIN?
+CME ERROR: 13 //(U)SIM failure.
AT+CFUN=1 //Switch UE to full functionality.
OK
AT+CPIN?
+CPIN: READY

OK
AT+COPS?
+COPS: 0,2,"46000",7 //Operator is registered.

OK
```

3 (U)SIM Related Commands

3.1. AT+CIMI Request International Mobile Subscriber Identity (IMSI)

This command requests the International Mobile Subscriber Identity (IMSI) which is intended to permit the TE to identify the individual SIM card or active application in the UICC (GSM or SIM) that is attached to MT.

AT+CIMI Request International Mobile Subscriber Identity (IMSI)	
Test Command AT+CIMI=?	Response OK
Execution Command AT+CIMI	Response <IMSI> OK If there is any error related to MT functionality: +CME ERROR: <err> Or ERROR
Maximum Response Time	300 ms
Characteristics	/
Reference 3GPP TS 27.007	

Parameter

<IMSI>	International Mobile Subscriber Identity (string without double quotes).
<err>	MT error result code. See Chapter 8.2 for details.

Example

AT+CIMI 460023210226023	//Query IMSI number of the (U)SIM attached to MT. //The IMSI number of (U)SIM attached to MT.
--	--

OK

3.2. AT+CLCK Facility Lock

This command locks, unlocks or interrogates an MT or a network facility **<fac>**. It can be aborted when network facilities are being set or queried. Password is normally needed to do such actions. The factory default password of PF, PN, PU, PP and PC lock is "12341234". When querying the status of a network service (**<mode>**=2), the response line for 'not active' case (**<status>**=0) should be returned only if the service is not active for any **<class>**.

AT+CLCK Facility Lock	
Test Command AT+CLCK=?	Response +CLCK: (list of supported <fac> s) OK
Write Command AT+CLCK=<fac>,<mode>[,<passwd>[,<class>]]	Response If <mode> does not equal 2 and the command is set successfully: OK If <mode> =2 and the command is set successfully: +CLCK: <status>[,<class>] [+CLCK: <status>[,<class>] [...] OK
Maximum Response Time	5 s
Characteristics	The command takes effect immediately. The configuration is saved automatically.
Reference <i>3GPP TS 27.007</i>	

Parameter

<fac>	String type. Facility lock type. "SC" (U)SIM (lock SIM/UICC card inserted in the currently selected card slot) (SIM/UICC asks password in MT power-up and when this lock command is issued). "AO" BAOC (Bar All Outgoing Calls) (see <i>3GPP TS 22.088 clause 1</i>) "OI" BOIC (Bar Outgoing International Calls) (see <i>3GPP TS 22.088 clause 1</i>)
--------------------	---

"OX"	BOIC-exHC (Bar Outgoing International Calls except to Home Country) (see 3GPP TS 22.088 clause 1)
"AI"	BAIC (Bar All Incoming Calls) (see 3GPP TS 22.088 clause 2)
"IR"	BIC-Roam (Bar Incoming Calls when Roaming outside the home country) (see 3GPP TS 22.088 clause 2)
"AB"	All Barring services (see 3GPP TS 22.030) (applicable only for <mode>=0)
"AG"	All outgoing barring services (see 3GPP TS 22.030) (applicable only for <mode>=0)
"AC"	All incoming barring services (see 3GPP TS 22.030) (applicable only for <mode>=0)
"FD"	SIM card or active application in the UICC (GSM or USIM) fixed dialing memory feature (if PIN2 authentication has not been done during the current session, PIN2 is required as <passwd>)
"PF"	Lock Phone to the very First inserted SIM/UICC card (also referred in the present document as PH-FSIM) (MT asks password when other SIM/UICC cards are inserted).
"PN"	Network Personalization (see 3GPP TS 22.022)
"PU"	Network Subset Personalization (see 3GPP TS 22.022)
"PP"	Service Provider Personalization (see 3GPP TS 22.022)
"PC"	Corporate Personalization (see 3GPP TS 22.022)
<mode>	Integer type. Operation mode.
0	Unlock
1	Lock
2	Query status
<passwd>	String type. Password.
<class>	Integer type. A sum of integers each representing a class of information.
1	Voice
2	Data
4	Fax
7	All telephony except SMS
8	Short message service
16	Data circuit synchronization
32	Data circuit asynchronization
<status>	Integer type. Current status of the specified <class> .
0	Not active
1	Active

Example

```

AT+CLCK="SC",2           //Query the status of (U)SIM card.
+CLCK: 0                 //The (U)SIM card is not active.

OK
AT+CLCK="SC",1,"1234"    //Lock (U)SIM card, and the password is 1234.

```

```

OK
AT+CLCK="SC",2           //Query the status of (U)SIM card.
+CLCK: 1                 //The (U)SIM card is active.

OK
AT+CLCK="SC",0,"1234"    //Unlock (U)SIM card, and the password is 1234.
OK

```

3.3. AT+CPIN Enter PIN

This command sends to the MT a password which is necessary before it can be operated ((U)SIM PIN, (U)SIM PUK, PH-SIM PIN, etc.), or queries whether the MT requires a password or not before it can be operated.

If the PIN is to be entered twice, the MT automatically repeats the PIN. If no PIN request is pending, no action is taken towards MT and an error message **+CME ERROR** is returned to TE.

If the PIN required is (U)SIM PUK or (U)SIM PUK2, the second parameter is required. This second PIN, **<newpin>**, is used to replace the old PIN in the (U)SIM.

AT+CPIN Enter PIN	
Test Command AT+CPIN=?	Response OK
Read Command AT+CPIN?	Response MT returns an alphanumeric string indicating whether or not a password is required. +CPIN: <code> OK If there is any error related to MT functionality: +CME ERROR: <err> Or ERROR
Write Command AT+CPIN=<pin>[,<newpin>]	Response OK
Maximum Response Time	5 s
Characteristics	The command takes effect immediately. The configuration is saved automatically.
Reference 3GPP TS 27.007	

Parameter

<code>	String without double quotes. Type of the password MT requires.
READY	MT is not pending for any password
SIM PIN	MT is waiting for (U)SIM PIN to be given
SIM PUK	MT is waiting for (U)SIM PUK to be given
SIM PIN2	MT is waiting for (U)SIM PIN2 to be given
SIM PUK2	MT is waiting for (U)SIM PUK2 to be given
PH-NET PIN	MT is waiting for network personalization password to be given
PH-NET PUK	MT is waiting for network personalization unlocking password to be given
PH-NETSUB PIN	MT is waiting for network subset personalization password to be given
PH-NETSUB PUK	MT is waiting for network subset personalization unlocking password to be given
PH-SP PIN	MT is waiting for service provider personalization password to be given
PH-SP PUK	MT is waiting for service provider personalization unlocking password to be given
PH-CORP PIN	MT is waiting for corporate personalization password to be given
PH-CORP PUK	MT is waiting for corporate personalization unlocking password to be given
<pin>	String type. Password. If the requested password was a PUK, such as (U)SIM PUK1, PH-FSIM PUK or another password, then <pin> must be followed by <newpin> .
<newpin>	String type. New password required if the requested code was a PUK.
<err>	MT error result code. See Chapter 8.2 for details.

Example

```
//Enter PIN.
AT+CPIN?
+CPIN: SIM PIN                                //Waiting (U)SIM PIN to be given.

OK
AT+CPIN="1234"                                //Enter PIN.
OK

+CPIN: READY
AT+CPIN?
+CPIN: READY                                //PIN has already been entered.

OK
//Enter PUK and PIN.
AT+CPIN?
```

```
+CPIN: SIM PUK //Waiting (U)SIM PUK to be given.

OK
AT+CPIN="26601934","1234" //Enter PUK and the new PIN password.
OK

+CPIN: READY
AT+CPIN?
+CPIN: READY //PUK has already been entered.

OK
```

3.4. AT+CPWD Change Password

The Write Command sets a new password for the facility lock function defined by **AT+CLCK**. The Test Command returns a list of pairs which present the available facilities and the maximum length of their passwords.

AT+CPWD Change Password	
Test Command AT+CPWD=?	Response +CPWD: list of supported (<fac>,<pwdlength>)s OK
Write Command AT+CPWD=<fac>,<oldpwd>,<newpwd>	Response OK
Maximum Response Time	5 s
Characteristics	The command takes effect immediately. The configuration is saved automatically.
Reference <i>3GPP TS 27.007</i>	

Parameter

<fac>		String type. Facility lock type.
"SC"	(U)SIM (lock SIM/UICC card) (SIM/UICC asks password in MT power-up and when this lock command is issued)	
"AO"	BAOC (Bar All Outgoing Calls, see <i>3GPP TS 22.088 clause 1</i>)	
"OI"	BOIC (Bar Outgoing International Calls, see <i>3GPP TS 22.088 clause 1</i>)	
"OX"	BOIC-exHC (Bar Outgoing International Calls except to Home Country, see <i>3GPP TS 22.088 clause 1</i>)	
"AI"	BAIC (Bar All Incoming Calls, see <i>3GPP TS 22.088 clause 2</i>)	

"IR"	BIC-Roam (Bar Incoming Calls when Roaming outside the home country, see 3GPP TS 22.088 clause 2)
"AB"	All barring services (see 3GPP TS 22.030, applicable only for <mode>=0)
"AG"	All outgoing barring services (see 3GPP TS 22.030, applicable only for <mode>=0)
"AC"	All incoming barring services (see 3GPP TS 22.030, applicable only for <mode>=0)
"P2"	(U)SIM PIN2
<pwdlength>	Integer type. Maximum length of password.
<oldpwd>	String type. Password specified for the facility from the user interface or with command.
<newpwd>	String type. New password.

Example

```

AT+CPIN?
+CPIN: READY

OK
AT+CPWD="SC","1234","4321"           //Change (U)SIM card password from "1234" to "4321".
OK
//Restart MT or re-activate the (U)SIM card.
AT+CPIN?                             //Waiting (U)SIM PIN to be given.
+CPIN: SIM PIN

OK
AT+CPIN="4321"                       //PIN must be entered to define a new password "4321".
OK
+CPIN: READY

```

3.5. AT+CSIM Generic (U)SIM Access

This command allows a direct control of the (U)SIM that is inserted in the currently selected card slot by a distant application on TE. TE should then keep the processing of (U)SIM information within the frame specified by GSM/UMTS.

AT+CSIM Generic (U)SIM Access	
Test Command AT+CSIM=?	Response OK
Write Command AT+CSIM=<length>,<command>	Response +CSIM: <length>,<response>

	OK Or ERROR If there is any error related to MT functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.
Reference	
3GPP TS 27.007	

Parameter

<length>	Length of the characters that are sent to TE in <command> or <response> (two times the actual length of the command or response).
<command>	Command passed on by the MT to the (U)SIM in the format as described in 3GPP TS 51.011 (hexadecimal character format).
<response>	Response to the command passed on by the (U)SIM to the MT in the format as described in 3GPP TS 51.011 (hexadecimal character format).
<err>	MT error result code. See Chapter 8.2 for details.

3.6. AT+CRSM Restricted (U)SIM Access

This command offers easier but more limited access to the (U)SIM database. It transmits to the MT the (U)SIM **<command>** and its required parameters.

AT+CRSM Restricted (U)SIM Access	
Test Command	Response
AT+CRSM=?	OK
Write Command	Response
AT+CRSM=<command>[,<fileid>[,<P1>,<P2>,<P3>[,<data>[,<pathid>]]]]	+CRSM: <sw1>,<sw2>[,<response>]
	OK Or ERROR If there is any error related to MT functionality: +CME ERROR: <err>

Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.
Reference	
3GPP TS 27.007	

Parameter

<command>	Integer type. (U)SIM command passed on by the MT to the (U)SIM. 176 READ BINARY 178 READ RECORD 192 GET RESPONSE 214 UPDATE BINARY 220 UPDATE RECORD 242 STATUS
<fileid>	Integer type. Identifier for an elementary data file on (U)SIM. Mandatory for every command except STATUS.
<P1>, <P2>, <P3>	Parameters transferred by the MT to the (U)SIM. These parameters are mandatory for every command, except GET RESPONSE and STATUS. The values are described in 3GPP TS 51.011.
<data>	Information which should be written to the (U)SIM (hexadecimal character format).
<pathid>	The directory path of an elementary file on the SIM/UICC in hexadecimal format as defined in ETSI TS 102 221.
<sw1>, <sw2>	Integer type. Information from the (U)SIM about the execution of the actual command. These parameters are delivered to the TE in both cases, on successful or failed execution of the command.
<response>	Response of a successful completion of the command previously issued (hexadecimal character format). STATUS and GET RESPONSE return data, which gives information about the current elementary data field. The information includes the type of file and its size (see 3GPP TS 51.011). After READ BINARY or READ RECORD command, the requested data is returned. <response> is not returned after a successful UPDATE BINARY or UPDATE RECORD command.
<err>	Error codes. For more details, refer to Chapter 8.2 .

4 Network Service Commands

4.1. AT+COPS PLMN Selection

This command returns the current operators and their status, and allows automatic or manual network selection.

The Test Command returns a set of five parameters, each representing an operator presenting in the network. The set consists of an integer indicating the availability of the operator **<stat>**, long and short alphanumeric format of the name of the operator, numeric format representation of the operator and access technology. Any of the formats may be unavailable and should then be an empty field. The list of operators shall be in the order of: home network, networks referenced in (U)SIM and other networks.

The Read Command returns the current mode and the currently selected operator. If no operator is selected, **<format>**, **<oper>** and **<AcT>** are omitted.

The Write Command forces an attempt to select and register to the GSM/UMTS network operator. If the selected operator is not available, no other operator shall be selected (except **<mode>=4**). The format of selected operator name shall apply to further Read Commands (**AT+COPS?**).

AT+COPS PLMN Selection	
Test Command AT+COPS=?	Response +COPS: [range of supported (<stat>),long alphanumeric <oper>,short alphanumeric <oper>,numeric <oper>,<AcT>)]s[,,(range of supported <mode>s),(range of supported <format>s)] OK If there is any error related to MT functionality: +CME ERROR: <err> Or ERROR
Read Command AT+COPS?	Response +COPS: <mode>[,<format>[,<oper>][,<AcT>]] OK

	If there is any error related to MT functionality: +CME ERROR: <err> Or ERROR
Write Command AT+COPS=<mode>[,<format>[,<oper>[,<AcT>]]]	Response OK If there is any error related to MT functionality: +CME ERROR: <err> Or ERROR
Maximum Response Time	180 s, determined by the network.
Characteristics	/
Reference	
3GPP TS 27.007	

Parameter

<stat>	Integer type. 0 Unknown 1 Operator available 2 Current operator 3 Operator forbidden
<oper>	String type. Operator in format as per <format> .
<mode>	Integer type. 0 Automatic (<oper> field is ignored) 1 Manual (<oper> field shall be present and <AcT> optionally) 2 Deregister from network 3 Set only <format> (for AT+COPS? Read Command), and do not attempt registration/deregistration (<oper> and <AcT> fields are ignored). This value is not applicable in the Read Command response. 4 Manual/automatic (<oper> field shall be presented). If manual selection fails, automatic mode (<mode>=0) is entered
<format>	Integer type. 0 Long format alphanumeric <oper> which can be up to 16 characters long 1 Short format alphanumeric <oper> 2 Numeric <oper>. GSM location area identification number
<AcT>	Integer type. Access technology selected. Values 3, 4, 5 and 6 occur only in the response of Read Command while MS is in data service state and is not intended for the AT+COPS Write Command. 0 GSM

2	UTRAN
3	GSM w/EGPRS
4	UTRAN w/HSDPA
5	UTRAN w/HSUPA
6	UTRAN w/HSDPA and HSUPA
7	E-UTRAN
<err>	MT error result code. See Chapter 8.2 for details.

Example

```

AT+COPS=?                                //List all current network operators.
+COPS: (1,"CHN-UNICOM","UNICOM","46001",2),(1,"CHN-UNICOM","UNICOM","46001",0),(2,"CH
N-UNICOM","UNICOM","46001",7),(1,"CHN-CT","CT","46011",7),(3,"CHINA MOBILE","CMCC","46
000",0),,(0-4),(0-2)

OK
AT+COPS?                                //Query the currently selected network operator.
+COPS: 0,0,"46000",7

OK

```

4.2. AT+CREG Network Registration Status

The Write Command controls the presentation of an unsolicited result code **+CREG: <stat>** when **<n>=1** and there is a change in the MT's circuit mode network registration status in GERAN/UTRAN/E-UTRAN, or unsolicited result code **+CREG: <stat>[,<lac>],[<ci>],[<AcT>]]** when **<n>=2** and there is a change of the network cell in GERAN/UTRAN/E-UTRAN.

The Read Command returns the status of result code presentation and an integer **<stat>** which shows whether the network has currently indicated the registration of the MT. Location information elements **<lac>**, **<ci>** and **<AcT>**, if available, are returned only when **<n>=2** and MT is registered in the network.

AT+CREG Network Registration Status	
Test Command AT+CREG=?	Response +CREG: (range of supported <n>s) OK
Read Command AT+CREG?	Response +CREG: <n>,<stat>[,<lac>,<ci>[,<AcT>]] OK

	If there is any error related to MT functionality: +CME ERROR: <err> Or ERROR
Write Command AT+CREG=[<n>]	Response OK
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.
Reference 3GPP TS 27.007	

Parameter

<n>	Integer type. Whether to enable network registration URC. 0 Disable network registration unsolicited result code 1 Enable network registration unsolicited result code: +CREG: <stat> 2 Enable network registration unsolicited result code with location information: +CREG: <stat>[,<lac>],[<ci>],[<AcT>]]
<stat>	Integer type. Circuit mode registration status. 0 Not registered. MT is not currently searching a new operator to register to 1 Registered, home network 2 Not registered, but MT is currently searching a new operator to register to 3 Registration denied 4 Unknown (e.g. out of GERAN/UTRAN/E-UTRAN coverage) 5 Registered, roaming
<lac>	String type. Two bytes location area code in hexadecimal format.
<ci>	String type. Four byte GERAN/UTRAN/E-UTRAN cell ID in hexadecimal format.
<AcT>	Integer type. Access technology of the serving cell. 0 GSM 2 UTRAN 3 GSM w/EGPRS 4 UTRAN w/HSDPA 5 UTRAN w/HSUPA 6 UTRAN w/HSDPA and HSUPA 7 E-UTRAN
<err>	MT error result code. See Chapter 8.2 for details.

Example

```
AT+CREG=2           //Activate extended URC mode.
OK
AT+CREG?
```

+CREG: 2,1,"550c","5a29c0b",7 //URC reporting MT has registered on an E-UTRAN network.

OK

4.3. AT+CSQ Signal Quality

The Execution Command returns received signal strength indication **<rssi>** and channel bit error rate **<ber>** from the MT.

AT+CSQ Signal Quality	
Test Command AT+CSQ=?	Response +CSQ: (list of supported <rssi>s),(list of supported <ber>s) OK
Execution Command AT+CSQ	Response +CSQ: <rssi>,<ber> OK If there is error related to MT functionality: +CME ERROR: <err> Or ERROR
Maximum Response Time	300 ms
Characteristics	/
Reference	
3GPP TS 27.007	

Parameter

<rssi>	Integer type. Received signal strength indication. 0 -113 dBm or less 1 -111 dBm 2–30 -109 dBm. to -53 dBm 31 -51 dBm or greater 99 Not known or not detectable
<ber>	Integer type. Channel bit error rate (in percent). 0–7 As RxQual values in the table in 3GPP TS 45.008 subclause 8.2.4 99 Not known or not detectable
<err>	MT error result code. See Chapter 8.2 for details.

Example

```
AT+CSQ=?
```

```
+CSQ: (0-31,99),(0-7,99)
```

```
OK
```

```
AT+CSQ
```

```
+CSQ: 28,99 //The received signal strength indicator is 28 and the channel bit error rate is 99.
```

```
OK
```

4.4. AT+CPOL Preferred PLMN List

This command edits the PLMN selector with Access Technology lists in the SIM card or active application in the UICC (GSM or USIM).

AT+CPOL Preferred PLMN List	
Test Command AT+CPOL=?	Response +CPOL: (list of supported <index>s),(range of supported <format>s) OK
Read Command AT+CPOL?	Response Query the list of preferred operators: +CPOL: <index1>,<format>,<oper1>[,<GSM_AcT1>,<GSM_Compact_AcT1>,<UTRAN_AcT1>,<E-UTRAN_AcT1>] [+CPOL: <index2>,<format>,<oper2>[,<GSM_AcT2>,<GSM_Compact_AcT2>,<UTRAN_AcT2>,<E-UTRAN_AcT2>] [...]] OK
Write Command AT+CPOL=<index>[,<format>[,<oper>[,<GSM_AcT>,<GSM_Compact_AcT>,<UTRAN_AcT>,<E-UTRAN_AcT>]]]	Response If the optional parameters are omitted, delete the operator with specified <index>: OK If any of the optional parameters is specified, edit the list of preferred operators: OK Or ERROR If there is any error related to MT functionality: +CME ERROR: <err>

Maximum Response Time	300 ms
Characteristics	/
Reference	
3GPP TS 27.007	

Parameter

<indexn>	Integer type. The order number of operator in the (U)SIM preferred operator list.
<format>	Integer type. 0 Long format alphanumeric <oper> 1 Short format alphanumeric <oper> 2 Numeric <oper>
<opern>	String type. <format> indicates the format is alphanumeric or numeric (see AT+COPS).
<GSM_AcTn>	Integer type. GSM access technology. 0 Access technology is not selected 1 Access technology is selected
<GSM_Compact_AcTn>	Integer type. GSM compact access technology. 0 Access technology is not selected 1 Access technology is selected
<UTRAN_AcTn>	Integer type. UTRAN access technology. 0 Access technology is not selected 1 Access technology is selected
<E-UTRAN_AcTn>	Integer type. E-UTRAN access technology. 0 Access technology is not selected 1 Access technology is selected
<err>	MT error result code. See Chapter 8.2 for details.

NOTE

The access technology selection parameters **<GSM_AcT>**, **<GSM_Compact_AcT>**, **<UTRAN_AcT>** and **<E-UTRAN_AcT>** are required for SIM cards or UICCs containing PLMN selector with access technology.

4.5. AT+COPN Read Operator Names

This command returns the list of operator names from the MT. Each operator code **<numericn>** that has an alphanumeric equivalent **<alphan>** in the MT memory is returned.

AT+COPN Read Operator Names	
Test Command AT+COPN=?	Response OK
Execution Command AT+COPN	Response +COPN: <numeric1>,<alpha1> [+COPN: <numeric2>,<alpha2> [...]] OK If there is any error related to MT functionality: +CME ERROR: <err>
Maximum Response Time	Depends on the number of operator names.
Characteristics	/
Reference 3GPP TS 27.007	

Parameter

<numericn>	String type. Operator names in numeric format (see AT+COPS).
<alphan>	String type. Operator names in long alphanumeric format (see AT+COPS).
<err>	MT error result code. See Chapter 8.2 for details.

4.6. AT+CTZU Automatic Time Zone Update

This Write Command enables and disables automatic time zone update via NITZ.

AT+CTZU Automatic Time Zone Update	
Test Command AT+CTZU=?	Response +CTZU: (list of supported <onoff>s) OK
Write Command AT+CTZU=<onoff>	Response OK Or ERROR
Read Command AT+CTZU?	Response +CTZU: <onoff>

	OK
Maximum Response Time	300 ms
Characteristic	/
Reference	
3GPP TS 27.007	

Parameter

<onoff>	Integer type. Enable or disable automatic time zone update.
0	Disable automatic time zone update via NITZ
<u>1</u>	Enable automatic time zone update via NITZ and update GMT time to URC

Example

AT+CTZU?

+CTZU: 1

OK

AT+CTZU=?

+CTZU: (0,1)

OK

AT+CTZU=0

OK

AT+CTZU?

+CTZU: 0

OK

4.7. AT+CTZR Time Zone Reporting

This command controls the time zone change event reporting. If reporting is enabled, the MT returns the unsolicited result code **+CTZV: <tz>** or **+CTZE: <tz>,<dst>,<time>** whenever the time zone is changed. The configuration is stored to flash automatically.

AT+CTZR Time Zone Reporting

Test Command	Response
AT+CTZR=?	+CTZR: (list of supported <reporting>s)

	OK
Write Command AT+CTZR=<reporting>	Response OK Or ERROR
Read Command AT+CTZR?	Response +CTZR: <reporting> OK
Maximum Response Time	300 ms
Characteristic	This command takes effect immediately. The configuration will be saved automatically.
Reference 3GPP TS 27.007	

Parameter

<reporting>	Integer type. Enable or disable time zone reporting. 0 Disable time zone change event reporting 1 Enable time zone change event reporting by URC +CTZV: <tz> 2 Enable extended time zone and local time reporting by URC +CTZE: <tz>,<dst>,<time>
<tz>	String type. Sum of the local time zone (difference between the local time and GMT is expressed in quarters of an hour) plus daylight saving time. The format is “±zz”, expressed as a fixed-width, two-digit integer from -48 to +56. To maintain a fixed width, numbers from -9 to +9 are expressed with a leading zero, e.g. “-09”, “+00” and “+09”.
<dst>	Integer type. Whether <tz> includes daylight savings adjustment. 0 <tz> includes no adjustment for daylight saving time 1 <tz> includes +1 hour (i.e. 4 quarters in <tz>) adjustment for daylight saving time 2 <tz> includes +2 hours (i.e. 8 quarters in <tz>) adjustment for daylight saving time
<time>	String type. Local time. The format is “YYYY/MM/DD,hh:mm:ss”, expressed as integers representing year (YYYY), month (MM), date (DD), hour (hh), minute (mm) and second (ss). This parameter can be provided by the network when delivering time zone information and be presented in the URC of extended time zone reporting.

5 Packet Domain Commands

5.1. AT+CGATT PS Attach or Detach

This command attaches the MT to, or detaches the MT from, the Packet Domain service. After the command has completed, the MT remains in V.250 command state. If MT is already in the requested state, the command is ignored and the **OK** response is returned. If the requested state cannot be achieved, an **ERROR** or **+CME ERROR** response is returned.

AT+CGATT PS Attach or Detach	
Test Command AT+CGATT=?	Response +CGATT: (list of supported <state>s) OK
Read Command AT+CGATT?	Response +CGATT: <state> OK
Write Command AT+CGATT=<state>	Response OK Or ERROR If there is any error related to MT functionality: +CME ERROR: <err>
Maximum Response Time	140 s, determined by the network.
Characteristics	/
Reference 3GPP TS 27.007	

Parameter

<state>	Integer type. The state of PS attachment. 0 Detached 1 Attached Other values are reserved and will result in an ERROR to the Write Command.
----------------------	---

<err> MT error result code. See **Chapter 8.2** for details.

Example

```
AT+CGATT=1           //Attach to PS service.
OK
AT+CGATT=0           //Detach from PS service.
OK
AT+CGATT?             //Query the current PS service state.
+CGATT: 0
OK
```

5.2. AT+CGDCONT Define PDP Context

This command specifies PDP context parameters for a specific context <cid>. A special form of the Write Command (**AT+CGDCONT=<cid>**) causes the values for context <cid> to become undefined. It is not allowed to change the definition of an already activated context.

This Read Command returns the current settings for each defined PDP context.

AT+CGDCONT Define PDP Context	
Test Command AT+CGDCONT=?	Response +CGDCONT: (range of supported <cid>s),<PDP_type>,<APN>,<PDP_addr>,(range of supported <d_comp>s),(range of supported <h_comp>s),(list of supported <IPv4AddrAlloc>s),(list of supported <request_type>s),(range of supported <P-CSCF_discovery>s),(list of supported <IM_CN_Signalling_Flag_Ind>s) OK
Read Command AT+CGDCONT?	Response +CGDCONT: <cid>,<PDP_type>,<APN>,<PDP_addr>,<d_comp>,<h_comp>,<IPv4AddrAlloc>,<request_type>,<P-CSCF_discovery>,<IM_CN_Signalling_Flag_Ind> [...] OK
Write Command AT+CGDCONT=<cid>[,<PDP_type>[,<APN>[,<PDP_addr>[,<d_comp>[,<h_c	Response OK Or

omp>[,<IPv4AddrAlloc>[,<request_type>[,<P-CSCF_discovery>[,<IM_CN_Signalling_Flag_Ind>]]]]]]]]	ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.
Reference	
3GPP TS 27.007	

Parameter

<cid>	Integer type. PDP context identifier. The parameter is local to the TE-MT interface and is used in other PDP context-related commands. The range of permitted values (minimum value=1) is returned by the Test Command. Range: 1–15.
<PDP_type>	String type. Packet data protocol type, a string parameter which specifies the type of packet data protocol. Only "IP", "IPv6" and "IPV4V6" are supported for EPS services. "IP" IPv4. Internet protocol (<i>IETF STD 5</i>) "PPP" Point to Point Protocol (<i>IETF STD 51</i>) "IPv6" Internet Protocol, version 6 (see <i>RFC 2460</i>) "IPV4V6" Virtual <PDP_type> introduced to handle dual IP stack UE capability (See <i>3GPP TS 24.301</i>)
<APN>	String type. Access point name, a logical name used to select the GGSN or the external packet data network. If the value is null or omitted, then the subscription value will be requested.
<PDP_addr>	String type. Identifies the MT in the address space applicable to the PDP. If the value is null or omitted, then a value may be provided by the TE during the PDP startup procedure or, failing that, a dynamic address will be requested. The allocated address may be read using AT+CGPADDR .
<d_comp>	Integer type. Controls PDP data compression (applicable for SNDCP only) (see <i>3GPP TS 44.065</i>). 0 Off 1 On (Manufacturer preferred compression) 2 V.42bis 3 V.44 (Not supported currently)
<h_comp>	Integer type. Controls PDP header compression (see <i>3GPP TS 44.065</i> and <i>3GPP TS 25.323</i>). 0 Off 1 On 2 RFC1144 [105] (applicable for SNDCP only) 3 RFC2507 [107]

<IPv4AddrAlloc>	4 RFC3095 [108] (applicable for PDCP only) Integer type. Controls how the MT/TA requests to get the IPv4 address information. 0 IPv4 address allocated through NAS signaling 1 IPv4 address allocated through DHCP
<request_type>	Integer type. Type of PDP context activation request for the PDP context. Refer to <i>3GPP TS 24.301 (subclause 6.5.1.2)</i> and <i>3GPP TS 24.008 (subclause 10.5.6.17)</i>. According to <i>3GPP TS 24.008 (subclause 4.2.4.2.2 and 4.2.5.1.4)</i> and <i>3GPP TS 24.301 (subclause 5.2.2.3.3 and 5.2.3.2.2)</i>, a separate PDP context must be established for emergency bearer services.
<P-CSCF_discovery>	Integer type. Influences how the MT/TA requests to get the P-CSCF address, (see <i>3GPP TS 24.229 annex B and L</i>). 0 Preference of P-CSCF address discovery not influenced by AT+CGDCONT 1 Preference of P-CSCF address discovery through NAS signaling 2 Preference of P-CSCF address discovery through DHCP
<IM_CN_Signalling_Flag_Ind>	Integer type. Indicates to the network whether the PDP context is for IM CN subsystem-related signaling only or not. 0 UE indicates that the PDP context is not for IM CN subsystem-related signaling only 1 UE indicates that the PDP context is for IM CN subsystem-related signaling only

5.3. AT+CGACT Activate or Deactivate PDP Contexts

This command activates or deactivates the specified PDP context(s) defined by **AT+CGDCONT**. After the command has completed, the MT remains in V.250 command state. If any PDP context is already in the requested state, the state for that context remains unchanged. If MT is not PS attached when the activation form of the command is executed, MT first performs a PS attach and then attempts to activate the specified contexts. If no **<cid>**s specify the activation/deactivation form of the command, the command activates or deactivates all defined PDP contexts.

AT+CGACT Activate or Deactivate PDP Contexts	
Test Command AT+CGACT=?	Response +CGACT: (list of supported <state>s) OK
Read Command AT+CGACT?	Response +CGACT: <cid>,<state> [...]

	OK
Write Command AT+CGACT=<state>,<cid>	Response OK Or NO CARRIER If there is any error related to MT functionality: +CME ERROR: <err> Or ERROR
Maximum Response Time	150 s, determined by the network.
Characteristics	/
Reference 3GPP TS 27.007	

Parameter

<state>	Integer type. State of PDP context activation. 0 Deactivated 1 Activated Other values are reserved and result in an ERROR response to the Write Command.
<cid>	Integer type. Specifies a particular PDP context definition (see AT+CGDCONT).
<err>	MT error result code. See Chapter 8.2 for details.

Example

```

AT+CGDCONT=2,"IP","UNINET"           //Define PDP context.
OK
AT+CGACT=1,2                           //Activate PDP context
OK
AT+CGACT?                               //Query the current PDP context state.
+C GACT: 1,1
+C GACT: 2,1

OK
AT+CGACT=0,2                           //Deactivate the PDP context
OK

```

5.4. AT+CGPADDR Show PDP Address(es)

This command returns a list of PDP addresses for the specified context identifiers. If no **<cid>** is specified, the addresses for all defined contexts are returned.

AT+CGPADDR Show PDP Address(es)	
Test Command AT+CGPADDR=?	Response +CGPADDR: (list of defined <cid>s) OK
Write Command AT+CGPADDR[=<cid>[,<cid>[,...]]]	Response +CGPADDR: <cid>,<PDP_addr> [...] OK Or ERROR
Maximum Response Time	300 ms
Characteristics	/
Reference	
3GPP TS 27.007	

Parameter

<cid>	Integer type. A particular PDP context definition (see AT+CGDCONT).
<PDP_addr>	String type. Identifies the MT in the address space applicable to the PDP. The address may be static or dynamic. For a static address, it is the one set by AT+CGDCONT when the context is defined. For a dynamic address, it is the one assigned during the last PDP context activation that used the context definition referred to by <cid> . <PDP_addr> is omitted if no address is available.

Example

```

AT+CGDCONT=1,"IP","UNINET"           //Define a PDP context.
OK
AT+CGACT=1,1                           //Activated the PDP.
OK
AT+CGPADDR=1                           //Show the PDP address.
+CGPADDR: 1,"10.76.51.180"
OK

```

5.5. AT+CGCLASS GPRS Mobile Station Class

This command sets the MT to operate according to the specified mode of operation. For details, see *3GPP TS 23.060*.

AT+CGCLASS GPRS Mobile Station Class	
Test Command AT+CGCLASS=?	Response +CGCLASS: (list of supported <class>s) OK
Read Command AT+CGCLASS?	Response +CGCLASS: <class> OK
Write Command AT+CGCLASS=<class>	Response OK If there is any error: +CME ERROR: <err> Or ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration will be saved automatically.
Reference <i>3GPP TS 27.007</i>	

Parameter

<class>	String type. GPRS mobile class (Functionality in descending order). A Class-A mode of operation (A/Gb mode), or CS/PS mode of operation (highest mode of operation) B Class-B mode of operation (A/Gb mode), or CS/PS mode of operation CG Class-C mode of operation in PS only mode (A/Gb mode), or PS mode of operation CC Class-C mode of operation in CS only mode (A/Gb mode), or CS (lowest mode of operation)
<err>	Error codes. For more details, refer to Chapter 8.2 .

5.6. AT+CGREG GPRS Network Registration Status

This command queries the network registration status and controls the presentation of an unsolicited result code **+CGREG: <stat>** when **<n>=1** and there is a change in the MT's GPRS network registration status in GERAN/UTRAN, or unsolicited result code **+CGREG: <stat>,[<lac>],[<ci>],[<AcT>],[<rac>]** when **<n>=2** and there is a change of the network cell in GERAN/UTRAN.

AT+CGREG GPRS Network Registration Status	
Test Command AT+CGREG=?	Response +CGREG: (range of supported <n>s) OK
Read Command AT+CGREG?	Response +CGREG: <n>,<stat>,[<lac>],[<ci>],[<AcT>],[<rac>] OK
Write Command AT+CGREG=<n>	Response OK Or ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.
Reference	3GPP TS 27.007

Parameter

<n>	Integer type. 0 Disable network registration unsolicited result code 1 Enable network registration unsolicited result code +CGREG: <stat> 2 Enable network registration and location information unsolicited result code +CGREG: <stat>,[<lac>],[<ci>],[<AcT>],[<rac>]
<stat>	Integer type. GPRS registration status. 0 Not registered. MT is not currently searching an operator to register to. The UE is in GMM state GMM-NUL or GMM-DEREGISTERED-INITIATED. The GPRS service is disabled, but the UE is allowed to attach for GPRS if requested by the user. 1 Registered, home network. The UE is in GMM state GMM-REGISTERED or GMM-ROUTING-AREA-UPDATING-INITIATED INITIATED on the home PLMN. 2 Not registered, but MT is trying to attach or searching an operator to register to. UE is in GMM state GMM-DEREGISTERED or GMM-REGISTERED-INITIATED. The GPRS service is enabled, but an allowable PLMN is currently not available. The UE will start

	a GPRS attach as soon as an allowable PLMN is available.
3	Registration denied. The UE is in GMM state GMM-NULL. The GPRS service is disabled, and the UE is not allowed to attach for GPRS if requested by the user.
4	Unknown (e.g. out of GERAN/UTRAN coverage)
5	Registered, roaming
<lac>	String type. Two-byte location area code in hexadecimal format (e.g. "00C3" equals 195 in decimal).
<ci>	String type. Four byte GERAN/UTRAN cell ID in hexadecimal format.
<AcT>	Integer type. Access technology selected.
0	GSM
2	UTRAN
3	GSM w/EGPRS
4	UTRAN w/HSDPA
5	UTRAN w/HSUPA
6	UTRAN w/HSDPA and HSUPA
7	E-UTRAN
<rac>	String type; one byte routing area code in hexadecimal format.

5.7. AT+CGEREP Packet Domain Event Reporting

This command enables or disables sending of unsolicited result codes **+CGEV: XXX** from MT to TE in the case of certain events occurring in the Packet Domain MT or the network. **<mode>** controls the processing of unsolicited result codes specified within this command. **<bfr>** controls the effect on buffered codes when **<mode>=1** or **2** is entered.

AT+CGEREP Packet Domain Event Reporting	
Test Command AT+CGEREP=?	Response +CGEREP: (range of supported <mode>s),(list of supported <bfr>s) OK
Read Command AT+CGEREP?	Response +CGEREP: <mode>,<bfr> OK Or ERROR
Write Command AT+CGEREP=<mode>[,<bfr>]	Response OK Or ERROR

Execution Command AT+CGEREP	Response OK
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations is saved automatically.
Reference 3GPP TS 27.007	

Parameter

<mode>	Integer type. 0 Buffer unsolicited result codes in the MT; if MT result code buffer is full, the oldest ones can be discarded. No codes are forwarded to the TE. 1 Discard unsolicited result codes when MT-TE link is reserved (e.g. in on-line data mode), otherwise forward them directly to the TE. 2 Buffer unsolicited result codes in the MT when MT-TE link is reserved (e.g. in on-line data mode) and flush them to the TE when MT-TE link becomes available. Otherwise forward them directly to the TE.
<bfr>	Integer type. 0 MT buffer of unsolicited result codes defined within this command is cleared when <mode> 1 or 2 is specified. 1 MT buffer of unsolicited result codes defined within this command is flushed to the TE when <mode> 1 or 2 is specified (OK response shall be given before flushing the codes).

NOTE

The unsolicited result codes and the corresponding events are defined as follows:

1. **+CGEV: REJECT <PDP_type>,<PDP_addr>**: A network request for PDP context activation occurred when the MT was unable to report it to the TE with a **+CRING** unsolicited result code and was automatically rejected. (See **AT+CGDCONT**)
Note: This event is not applicable for EPS.
2. **+CGEV: NW REACT <PDP_type>,<PDP_addr>,<cid>**: The network has requested a context reactivation. The **<cid>** used to reactivate the context is provided if known to the MT. (See **AT+CGDCONT**)
Note: This event is not applicable for EPS.
3. **+CGEV: NW DEACT <PDP_type>,<PDP_addr>,<cid>**: The network has forced a context deactivation. The **<cid>** used to activate the context is provided if known to the MT. (See **AT+CGDCONT**)
4. **+CGEV: ME DEACT <PDP_type>,<PDP_addr>,<cid>**: The mobile equipment has forced a context deactivation. The **<cid>** used to activate the context is provided if known to the MT. (See **AT+CGDCONT**)

5. **+CGEV: NW DETACH**: The network has forced a Packet Domain detach. This implies that all active contexts have been deactivated. These are not reported separately.
6. **+CGEV: ME DETACH**: The mobile equipment has forced a Packet Domain detach. This implies that all active contexts have been deactivated. These are not reported separately.
7. **+CGEV: NW CLASS <class>**: The network has forced a change of MS class. The highest available class is reported (see **AT+CGCLASS**).
8. **+CGEV: ME CLASS <class>**: The mobile equipment has forced a change of MS class. The highest available class is reported (see **AT+CGCLASS**).
9. **+CGEV: PDN ACT <cid>**: Activated a context. The context represents a PDN connection in LTE or a Primary PDP context in GSM/UMTS. (See **AT+CGDCONT**)
10. **+CGEV: PDN DEACT <cid>**: Deactivated a context. The context represents a PDN connection in LTE or a Primary PDP context in GSM/UMTS. (See **AT+CGDCONT**)

5.8. AT+CGSMS Select Service for MO SMS Messages

This command specifies the service or service preference that the MT will use to send MO (mobile originated) SMS messages.

AT+CGSMS Select Service for MO SMS Messages	
Test Command AT+CGSMS=?	Response +CGSMS: (range of currently available <service>s) OK
Read Command AT+CGSMS?	Response +CGSMS: <service> OK
Write Command AT+CGSMS=<service>	Response OK If there is any error related to MT functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.
Reference	3GPP TS 27.007

Parameter

<service>	Integer type. Service or service preference to be used. 0 Packet domain <u>1</u> Circuit switched 2 Packet domain preferred (use circuit switched if GPRS not available) 3 Circuit switch preferred (use Packet domain if circuit switched not available)
<err>	MT error result code. See Chapter 8.2 for details.

5.9. AT+CEREG EPS Network Registration Status

This command queries the network registration status and controls the presentation of an unsolicited result code **+CEREG: <stat>** when **<n>=1** and there is a change in the MT's EPS network registration status in E-UTRAN, or unsolicited result code **+CEREG: <stat>[,<tac>],[<ci>],[<AcT>]]** when **<n>=2** and there is a change of the network cell in E-UTRAN.

AT+CEREG EPS Network Registration Status	
Test Command AT+CEREG=?	Response +CEREG: (range of supported <n>s) OK
Read Command AT+CEREG?	Response +CEREG: <n>,<stat>[,<tac>,<ci>[,<AcT>]] OK
Write Command AT+CEREG=[<n>]	Response OK Or ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.
Reference 3GPP TS 27.007	

Parameter

<n>	Integer type. <u>0</u> Disable network registration unsolicited result code 1 Enable network registration unsolicited result code +CEREG: <stat>
------------------	---

	2	Enable network registration and location information unsolicited result code +CEREG: <stat>[,<tac>],[<ci>],[<AcT>]]
<stat>		Integer type. EPS registration status.
	0	Not registered. MT is not currently searching an operator to register to
	1	Registered, home network
	2	Not registered, but MT is currently trying to attach or searching an operator to register to
	3	Registration denied
	4	Unknown (e.g. out of E-UTRAN coverage)
	5	Registered, roaming
<tac>		String type. Two-byte tracking area code in hexadecimal format.
<ci>		String type. Four byte E-UTRAN cell ID in hexadecimal format.
<AcT>		Integer type. Access technology selected.
	0	GSM
	2	UTRAN
	3	GSM w/EGPRS
	4	UTRAN w/HSDPA
	5	UTRAN w/HSUPA
	6	UTRAN w/HSDPA and HSUPA
	7	E-UTRAN

6 Audio Commands

6.1. AT+CLVL Loudspeaker Volume Level

This command selects the volume level of the internal loudspeaker of the MT.

AT+CLVL Loudspeaker Volume Level	
Test Command AT+CLVL=?	Response +CLVL: (list of supported<level>s) OK
Read Command AT+CLVL?	Response +CLVL: <level> OK Or ERROR
Write Command AT+CLVL=<level>	Response OK Or ERROR If there is any error related to MT functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.
Reference Reference 3GPP TS 27.007	

Parameter

<level>	Integer type. Volume level with manufacturer specific range (Smallest value represents the lowest sound level). Range: 0–5; Default: 3.
<err>	MT error result code. See Chapter 8.2 for details.

6.2. AT+VTS DTMF and Tone Generation

This command sends ASCII characters which cause MSC to transmit DTMF tones to a remote subscriber. This command can only be operated in a voice call.

AT+VTS DTMF and Tone Generation	
Test Command AT+VTS=?	Response +VTS: (list of supported <DTMF_string>s),(range of supported <duration>s) OK
Write Command AT+VTS=<DTMF_string>[,<duration>]	Response OK Or ERROR If there is any error related to MT functionality: +CME ERROR: <err>
Maximum Response Time	Depends on the length of <DTMF_string> and <duration> .
Characteristics	/
Reference <i>3GPP TS 27.007</i>	

Parameter

<DTMF_string>	String type. ASCII characters in the set 0...9,#,*, A, B, C, D. The string should be enclosed in quotation marks ("..."). When sending multiple tones at a time, the time interval of two tones <interval> can be specified by AT+VTD . The maximal length of the string is 31 bytes.
<duration>	The duration of each tone in 10 ms with tolerance. Range: 300–600. Default: 300. If the duration is less than the minimum time specified by the network, the actual duration will be the network specified time. If this parameter is omitted, <duration> can be specified by AT+VTD .
<err>	MT error result code. See Chapter 8.2 for details.

Example

```

ATD12345678900;           //Dial.
OK
//Call connected.
AT+VTS="1"                //The remote caller can hear the DTMF tone.
OK

```

```
AT+VTS="1234567890A" //Send multiple tones at a time.
OK
```

6.3. AT+VTD Tone Duration

This command sets the duration of DTMF tones. It can also set time interval of two tones when sending multiple tones at a time.

AT+VTD Tone Duration	
Test Command AT+VTD=?	Response +VTD: (range of supported <duration>s),(range of supported <interval>s) OK
Read Command AT+VTD?	Response +VTD: <duration>,<interval> OK
Write Command AT+VTD=<duration>[,<interval>]	Response OK Or ERROR If there is any error related to MT functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.
Reference 3GPP TS 27.007	

Parameter

<duration>	Integer type. The duration tone in 1/10 seconds with tolerance. Range: 300-600. Default value: 300. If the duration is less than the minimum time specified by the network, the actual duration will be network specified time.
<interval>	Integer type. The time interval of two tones when sending multiple tones at a time by AT+VTS . Range: 300-600. Default value: 300. Unit: 100 ms.
<err>	MT error result code. See Chapter 8.2 for details.

7 Hardware Related Commands

7.1. AT+QADC Read ADC Value

This command reads the voltage value of ADC channel.

AT+QADC Read ADC Value	
Test Command AT+QADC=?	Response +QADC: (list of supported <port>s) OK
Read Command AT+QADC=<port>	Response +QADC: <status>,<value> OK
Maximum Response Time	300 ms
Characteristics	/

Parameter

<port>	Integer type. Channel number of the ADC. 0 ADC channel 0 1 ADC channel 1
<status>	Integer type. Whether the ADC value read is successful. 0 Failed 1 Successful
<value>	Integer type. The voltage of specified ADC channel. Unit: mV.

8 Appendix References

8.1. References

Table 2: Terms and Abbreviations

Abbreviation	Description
3GPP	3rd Generation Partnership Project
ADC	Analog-to-Digital Converter
APN	Access Point Name
ASCII	American Standard Code for Information Interchange
BAIC	Bar All Incoming Calls
BAOC	Bar All Outgoing Calls
BCD	Binary-Coded Decimal
BOIC	Bar Outgoing International Calls
BOIC-exHC	Bar Outgoing International Calls except to Home Country
CS	Circuit Switching
DCE	Data Communication Equipment
DHCP	Dynamic Host Configuration Protocol
DTE	Data Terminal Equipment
DTMF	Dual-Tone Multifrequency
EGPRS	Enhanced General Packet Radio Service
EPS	Evolved Packet System

E-UTRAN	Evolved Universal Terrestrial Radio Access Network
GGSN	Gateway GPRS Support Node
GMM	GPRS Mobility Management
GMT	Greenwich Mean Time
GPRS	General Packet Radio Service
GSM	Global System for Mobile Communications
HSDPA	High Speed Downlink Packet Access
HSUPA	High Speed Uplink Packet Access
IMEI	International Mobile Equipment Identity
IMSI	International Mobile Subscriber Identity
IP	Internet Protocol
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
IRA	International Reference Alphabet
ME	Mobile Equipment
MO	Mobile Originated
MS	Mobile Station
MSC	Mobile Switching Center
MT	Mobile Terminal
NAS	Non-Access Stratum
NITZ	Network Identity and Time Zone/Network Informed Time Zone
P-CSCF	Proxy-Call Session Control Function
PDP	Packet Data Protocol
PDU	Packet Data Unit
PIN	Personal Identification Number

PLMN	Public Land Mobile Network
PPP	Point-to-Point Protocol
PS	Packet Switching
PUK	PIN Unlock Key
RAT	Radio Access Technology
RF	Radio Frequency
RX	Receive
RxQual	Receive Quality
SM	Session Management
SMS	Short Message Service
SMSC	Short Message Service Center
SNDCP	SubNetwork Dependent Convergence Protocol
TA	Terminal Adapter
TE	Terminal Equipment
TFT	Traffic Flow Template
TPDU	Transport Protocol Data Unit
UART	Universal Asynchronous Receiver/Transmitter
USC-2	Universal Character Set (UCS-2) Format
UE	User Equipment
UICC	Universal Integrated Circuit Card
UIM	User Identity Model
UMTS	Universal Mobile Telecommunications System
URC	Unsolicited Result Code
USB	Universal Serial Bus
(U)SIM	(Universal) Subscriber Identity Module

VLR

Visiting Location Register

8.2. Mobile Termination Error Result Code +CME ERROR

Final result code **+CME ERROR: <err>** indicates an error related to mobile equipment or network. The operation of **+CME ERROR: <err>** final result code is similar to the regular **ERROR** result code: if **+CME ERROR: <err>** is the result code for any of the commands in a command line, none of the following commands in the same command line is executed (neither **ERROR** nor **OK** result code shall be returned as a result of a completed command line execution). The format of **<err>** can be either numeric or verbose.

The following table lists some general errors. For some GSM protocol failure cause described in GSM specifications, the corresponding **ERROR** codes are not included.

Table 3: Summary of General +CME ERROR Codes

Numeric <err> Value	Verbose <err> Value
0	phone failure
1	no connection to phone
2	phone-adaptor link reserved
3	operation not allowed
4	operation not supported
5	PH-SIM PIN required
6	PH-FSIM PIN required
7	PH-FSIM PUK required
10	SIM not inserted
11	SIM PIN required
12	SIM PUK required
13	SIM failure
14	SIM busy

15	SIM wrong
16	incorrect password
17	SIM PIN2 required
18	SIM PUK2 required
20	memory full
21	invalid index
22	not found
23	memory failure
24	text string too long
25	invalid characters in text string
26	dial string too long
27	invalid characters in dial string
30	no network service
31	network timeout
32	network not allowed - emergency calls only
40	network personalization PIN required
41	network personalization PUK required
42	network subset personalization PIN required
43	network subset personalization PUK required
44	service provider personalization PIN required
45	service provider personalization PUK required
46	corporate personalization PIN required
47	corporate personalization PUK required
901	audio unknown error
902	audio invalid parameters

903	audio operation not supported
904	audio device busy

8.3. Summary of URC

Table 4: Summary of URC

Index	URC Display	Meaning	Condition
1	+CREG: <stat>	Registration status of the MT	AT+CREG=1
2	+CREG: <stat>[,<lac>],[<ci>],[<AcT>]]	After cell neighborhood changing shows whether the network has currently indicated the registration of the MT, with location area code	AT+CREG=2
3	+CGREG: <stat>	Network registration status of the MT	AT+CGREG=1
4	+CGREG: <stat>[,<lac>],[<ci>],[<AcT>],[<rac>]]	Network registration and location information of the MT	AT+CGREG=2
5	+CEREG: <stat>	EPS network registration status of the MT	AT+CEREG=1
6	+CEREG: <stat>[,<tac>],[<ci>],[<AcT>]]	EPS network registration and location information of the MT	AT+CEREG=2
7	+CTZV: <tz>	Time zone reporting	AT+CTZR=1
8	+CTZE: <tz>,<dst>,<time>	Extended time zone reporting	AT+CTZR=2
9	RING	An incoming call	N/A
10	+CFUN: 1	All function of the MT is available	N/A
11	+CPIN: <state>	(U)SIM card pin state	N/A
12	POWERED DOWN	Module power down	AT+QPOWD
13	+CGEV: REJECT <PDP_type>,<PDP_addr>	A network request for PDP activation, and was automatically rejected.	AT+CGEREP=2,1
14	+CGEV: NW REACT <PDP_type>,<PDP_addr>,<cid>]	The network request PDP reactivation	AT+CGEREP=2,1
15	+CGEV: NW DEACT <PDP_type>,<PDP_addr>,<cid>]	The network has forced a context deactivation	AT+CGEREP=2,1

16	+CGEV: ME DEACT <PDP_type>,<PDP_addr>,[<cid>]	The ME has forced a context deactivation.	AT+CGEREP=2,1
17	+CGEV: NW DETACH	The network has forced a Packet Domain detach.	AT+CGEREP=2,1
18	+CGEV: ME DETACH	The mobile equipment has forced a Packet Domain detach.	AT+CGEREP=2,1
19	+CGEV: NW CLASS <class>	The network has forced a change of MS class.	AT+CGEREP=2,1
20	+CGEV: ME CLASS <class>	The mobile equipment has forced a change of MS class.	AT+CGEREP=2,1
21	+CGEV: PDN ACT <cid>	Activated a context.	AT+CGEREP=2,1
22	+CGEV: PDN DEACT <cid>	Deactivated a context.	AT+CGEREP=2,1